

JAVASCRIPT ESSENTIALS

FUNCTIONS

- A FUNCTION IN JAVASCRIPT IS A REUSABLE BLOCK OF CODE DESIGNED TO PERFORM A SPECIFIC TASK. IT IS EXECUTED WHEN IT IS INVOKED (CALLED)
- IN JAVASCRIPT, FUNCTIONS ARE **FIRST-CLASS OBJECTS**. THIS MEANS THEY CAN BE STORED IN VARIABLES, PASSED AS ARGUMENTS TO OTHER FUNCTIONS, AND RETURNED FROM FUNCTIONS JUST LIKE ANY OTHER VALUE (LIKE STRINGS OR NUMBERS).

FUNCTION DECLARATION (NAMED)

- THE TRADITIONAL WAY TO DEFINE A FUNCTION. IT USES THE **FUNCTION** KEYWORD FOLLOWED BY A NAME.
- THESE ARE HOISTED, MEANING THEY CAN BE CALLED BEFORE THEY ARE DEFINED IN THE CODE.

FUNCTION EXPRESSION

- A FUNCTION DEFINED INSIDE AN EXPRESSION AND ASSIGNED TO A VARIABLE.
- THESE ARE NOT HOISTED, SO YOU MUST DEFINE THEM BEFORE CALLING THEM.
- THEY CAN BE ANONYMOUS (NO NAME)

ARROW FUNCTIONS

- INTRODUCED IN ES6, THESE PROVIDE A SHORTER SYNTAX. THEY DO NOT HAVE THEIR OWN CONTEXT FOR THE THIS KEYWORD, MAKING THEM IDEAL FOR CALLBACKS AND ARRAY METHODS.

IMMEDIATELY INVOKED FUNCTION EXPRESSIONS (IIFE)

A FUNCTION THAT RUNS AS SOON AS IT IS DEFINED.

IT IS WRAPPED IN PARENTHESES TO CREATE A PRIVATE SCOPE, PREVENTING VARIABLE POLLUTION IN THE GLOBAL ENVIRONMENT.

SCOPE AND THE SCOPE CHAIN

- SCOPE
 - JAVASCRIPT HAS THREE SCOPE LEVELS: GLOBAL SCOPE (ACCESSIBLE EVERYWHERE), FUNCTION SCOPE (VARIABLES DECLARED WITH VAR INSIDE A FUNCTION), AND BLOCK SCOPE (VARIABLES DECLARED WITH LET/CONST INSIDE {})
- SCOPE CHAIN
 - WHEN JAVASCRIPT LOOKS UP A VARIABLE, IT STARTS IN THE CURRENT SCOPE AND WALKS UP THE CHAIN OF ENCLOSING SCOPES UNTIL IT FINDS THE VARIABLE OR REACHES THE GLOBAL SCOPE.

WHERE TO USE

Scenario	Best choice	Why
App core logic	Function Declaration	Predictable, hoisted
Helper utilities	Function Declaration	Readable and reusable
Event handlers	Function Expression	Passed as value
Array methods	Arrow Function	Short and expressive
Conditional logic	Function Expression	Dynamic assignment
One-time setup	IIFE	Runs once
Private variables	IIFE	Encapsulation

CLOSURES

- A CLOSURE IS A FUNCTION THAT RETAINS ACCESS TO VARIABLES FROM ITS ENCLOSING SCOPE EVEN AFTER THE OUTER FUNCTION HAS RETURNED. EVERY FUNCTION IN JAVASCRIPT FORMS A CLOSURE.
- DATA ENCAPSULATION
 - CLOSURES ENABLE PRIVATE STATE — VARIABLES THAT CANNOT BE ACCESSED DIRECTLY FROM OUTSIDE THE FUNCTION, ONLY THROUGH THE RETURNED INTERFACE.

HIGHER-ORDER FUNCTIONS

- WHEN A FUNCTION RECEIVES ANOTHER FUNCTION AS AN ARGUMENT, IT IS KNOWN AS A HIGHER-ORDER FUNCTION, AND THE FUNCTION THAT IS PASSED AS AN ARGUMENT IS CALLED A CALLBACK.
- COMPOSITION COMBINES SIMPLE FUNCTIONS TO BUILD COMPLEX BEHAVIOR. THE OUTPUT OF ONE FUNCTION BECOMES THE INPUT OF THE NEXT.
- CURRYING TRANSFORMS A FUNCTION WITH MULTIPLE ARGUMENTS INTO A SEQUENCE OF FUNCTIONS, EACH TAKING A SINGLE ARGUMENT.

EJERCICIO 04

- ESCRIBIR UNA FUNCION QUE ORDENE UN ARREGLO, USANDO UNA FUNCION DE COMPARACION DINAMICA
- EJEMPLO:

```
const numeros = [3, 1, 4, 2];  
  
const ordenadoAsc = ordenar(numeros, (a, b) => a - b);  
const ordenadoDesc = ordenar(numeros, (a, b) => b - a);  
  
console.log(ordenadoAsc); // [1,2,3,4]  
console.log(ordenadoDesc); // [4,3,2,1]
```

OBJECT

- JAVASCRIPT OBJECT IS A STANDALONE ENTITY WITH PROPERTIES AND TYPE. IT CAN BE DEFINED AS AN UNORDERED COLLECTION OF RELATED DATA STORED AS KEY-VALUE PAIRS.
- LITERAL
 - `CONST USER = { NAME: "ALICE" };`
- CONSTRUCTOR (CLASS, NEW KEYWORD)
 - `CONST USER = NEW CLASSNAME();`

PROPERTIES AND METHODS

- PROPERTY: A VARIABLE ATTACHED TO AN OBJECT THAT REPRESENTS ITS CHARACTERISTICS (STATE).
- METHOD: A FUNCTION ATTACHED TO AN OBJECT THAT REPRESENTS ITS BEHAVIOR (ACTIONS).
- PROPERTY SHORTHAND: IF THE VARIABLE NAME MATCHES THE OBJECT KEY NAME, YOU CAN OMIT THE VALUE (CONST NAME = "ALICE"; CONST USER = { NAME };

PROPERTY ACCESSORS

- DOT NOTATION (OBJ.KEY): CLEAN AND STANDARD. USED WHEN YOU KNOW THE PROPERTY NAME BEFOREHAND.
- BRACKET NOTATION (OBJ["KEY"]): DYNAMIC. NECESSARY WHEN THE PROPERTY NAME CONTAINS SPACES, SPECIAL CHARACTERS, OR IS STORED INSIDE A VARIABLE.

OBJECT DESTRUCTURING

- BASIC DESTRUCTURING
 - DESTRUCTURING EXTRACTS VALUES FROM OBJECTS INTO DISTINCT VARIABLES USING A SYNTAX THAT MIRRORS OBJECT LITERAL STRUCTURE.
- ADVANCED PATTERNS
 - NESTED DESTRUCTURING REACHES INTO DEEPLY NESTED OBJECTS. DEFAULT VALUES PROVIDE FALLBACKS. RENAMING MAPS OBJECT KEYS TO DIFFERENT VARIABLE NAMES.

REFERENCE TYPE (THE POINTER CONCEPT)

- OBJECTS ARE NOT STORED BY VALUE; THEY ARE STORED BY REFERENCE IN MEMORY.
- MODIFYING B WILL AUTOMATICALLY ALTER A.
- TWO IDENTICAL-LOOKING OBJECTS ARE NEVER EQUAL: `{}` `===` `{}` RESULTS IN FALSE.

ARRAY METHODS

map

filter

forEach

find

some

every

reduce

SPREAD AND REST OPERATORS/PARAMETERS (...)

- SPREAD OPERATOR
 - EXPANDS AN ITERABLE (ARRAY, STRING) INTO INDIVIDUAL ELEMENTS. FOR OBJECTS, IT COPIES OWN ENUMERABLE PROPERTIES.
 - SPREAD CREATES SHALLOW COPIES — NESTED OBJECTS ARE STILL SHARED BY REFERENCE. USE IT FOR IMMUTABLE UPDATE PATTERNS.
- REST PARAMETERS
 - THE REST PARAMETER SYNTAX ALLOWS A FUNCTION TO ACCEPT AN INDEFINITE NUMBER OF ARGUMENTS AS A SINGLE ARRAY
- REST OPERATOR
 - GATHER OR "COLLECT" MULTIPLE ELEMENTS INTO A SINGLE VARIABLE. IT IS MOST COMMONLY USED IN DESTRUCTURING ASSIGNMENTS

JSON PARSING AND SERIALIZATION

- `JSON.stringify`
 - CONVERTS A JAVASCRIPT VALUE TO A JSON STRING. IT ACCEPTS OPTIONAL REPLACER AND SPACE ARGUMENTS FOR FILTERING AND FORMATTING.
- `JSON.parse`
 - PARSES A JSON STRING AND RETURNS THE CORRESPONDING JAVASCRIPT VALUE. A REVIVER FUNCTION CAN TRANSFORM VALUES DURING PARSING.

CLASSES AND PROTOTYPES

- ES6 CLASSES PROVIDE A CLEANER SYNTAX FOR CONSTRUCTOR FUNCTIONS AND PROTOTYPAL INHERITANCE. UNDER THE HOOD, THEY STILL USE PROTOTYPES.
- THE ``EXTENDS`` KEYWORD CREATES A SUBCLASS. THE ``SUPER`` KEYWORD CALLS THE PARENT CONSTRUCTOR AND METHODS.
- ABSTRACTION AND ENCAPSULATION
- POLYMORPHISM
- GET/SET

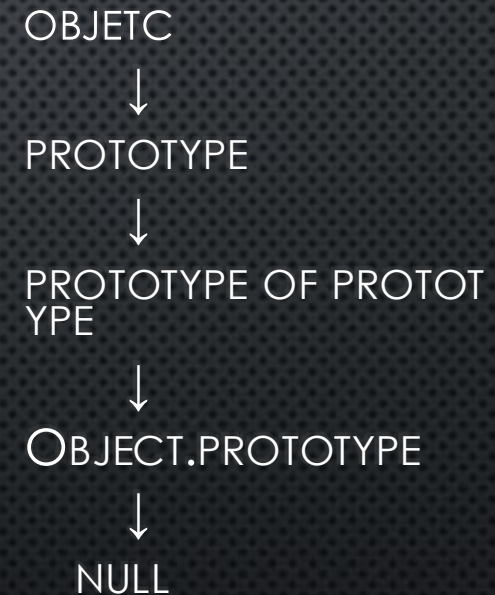
NO POSSIBLE WITH JS

- INTERFACES
- ABSTRACT CLASSES
- NO TRADITIONAL METHOD OVERLOADING
- STRICT TYPING

PROTOTYPE AND PROTOTYPE CHAIN

Prototype is a built-in mechanism by which JavaScript objects **inherit features and methods from one another**

Every object in JavaScript has an internal link to another object called its prototype. This creates a chain of objects known as the Prototype Chain



EJERCICIO 05

- UTILIZAR LA INFORMACION DEL ARCHIVO COMPRAS_CLIENTES.JSON PARA GENERAR UN REPORTE CON EL NOMBRE DEL CLIENTE, TOTAL GASTADO, TOTAL PRODUCTOS, EL PRODUCTO MÁS CARO Y CUÁNTO SE AHORRÓ EN DESCUENTOS, LA RESPUESTA DEBE ESTAR EN FORMATO JSON

```
[
  {
    "cliente": "Ana García",
    "totalProductos": 8,
    "subtotal": 15930,
    "totalAhorrado": 3784.50,
    "totalFinal": 12145.50,
    "productoMasCaro": "Laptop"
  },
  {
    "cliente": "Jose Mendoza",
    "totalProductos": 2,
    "subtotal": 600,
    "totalAhorrado": 0,
    "totalFinal": 600,
    "productoMasCaro": "Mesa"
  }
]
```